# ME 5309 – Mid-Semester Examination: ANSWER SHEET

**Roll No.:** ME25M025 **Name:** Ritika Chopra

**Questions attempted:** Q2, Q3, Q4

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## Question 2

*Given: $J = 0.02$ kg·m², $b = 0.05$ N·m·s/rad, $m = 1$ kg, $\ell = 0.25$ m, $g = 9.81$ m/s². So $mg\ell = 1 \times 9.81 \times 0.25 = 2.4525$ N·m (used throughout).*

**(a)** The model is second order, so I choose the position and velocity as states: $x = [x_1, x_2]^T = [\theta, \dot{\theta}]^T$, with input $\tau$.

Rewriting $J\ddot{\theta} + b\dot{\theta} + mg\ell \sin\theta = \tau$ gives the nonlinear state-space model: $\dot{x}_1 = x_2$, $\dot{x}_2 = (1/J)(\tau - b\,x_2 - mg\ell \sin x_1)$, i.e. $\dot{x} = f(x, \tau)$.

Equilibira for constant $\tau_0$ need $\dot{x} = 0$: the first equation gives $x_2 = 0$, and the second then gives $mg\ell \sin\theta_e = \tau_0$, i.e. $\sin\theta_e = \tau_0/(mg\ell)$.

This has a solution only if $|\tau_0/(mg\ell)| \le 1$, so $\mathbf{|\tau_0| \le mg\ell = 2.4525\ N{\cdot}m}$. For $|\tau_0| < mg\ell$ there are two equilibria per revolution ($\theta_e$ and $\pi - \theta_e$); at $|\tau_0| = mg\ell$ they merge, and beyond it the joint cannot be held statically.

**(b)** The equilibrium at $\theta_e = 60°$ needs the torque $\tau_e = mg\ell \sin 60° = 2.4525 \times 0.8660 = 2.124$ N·m. Define deviations $\delta x = x - x_e$ with $x_e = [60°, 0]^T$ and $\delta\tau = \tau - \tau_e$; then $\delta\dot{x} \approx A\,\delta x + B\,\delta\tau$ with the Jacobians evaluated at the equilibrium.

$\partial f_1/\partial x = [0, 1]$. $\partial f_2/\partial x_1 = -(mg\ell/J)\cos x_1 = -(2.4525/0.02)(0.5) = -122.625 \times 0.5 = -61.31$ ($s^{-2}$). $\partial f_2/\partial x_2 = -b/J = -0.05/0.02 = -2.5\ s^{-1}$.

**$A = [0, 1; -61.31, -2.5]$**, and since $\partial f_2/\partial\tau = 1/J = 50$, **$B = [0; 50]$**.

Check: the characteristic equation $\lambda^2 + 2.5\lambda + 61.31 = 0$ gives $\lambda = -1.25 \pm 7.73j$, a lightly damped stable oscillation, as expected for a pendulum hanging near-vertical (60° below the downward vertical).

**(c)** Set all time derivatives to zero in the closed loop. With $\tau = K_p(\theta_d - \theta) - K_d\dot{\theta}$ and $\dot{\theta} = \ddot{\theta} = 0$ the damper term vanishes (so $K_d$ has no effect on the final position) and the dynamics reduce to $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss})$.

Why $\theta_{ss} \neq \theta_d$: if the joint were at rest exactly at $\theta_d = 90°$ the PD torqe would be $K_p{\cdot}0 - K_d{\cdot}0 = 0$, yet gravity pulls with $mg\ell \sin 90° = 2.4525$ N·m. The net torque would be non-zero, so the joint could not be at rest. A purely proportional-derivative law can only hold a load if there is a steady error.

Let $e = \theta_d - \theta_{ss}$ (in rad). Then $\sin\theta_{ss} = \sin(\pi/2 - e) = \cos e$, and the equation becomes $50\,e = 2.4525 \cos e$. Fixed-point iteration: $e_0 = 2.4525/50 = 0.04905$ (taking $\cos e \approx 1$); $e_1 = 0.04905 \times \cos(0.04905) = 0.04899$; $e_2 = 0.04905 \times \cos(0.04899) = 0.04899$, converged.

$e = 0.04899$ rad $\times (180/\pi) = 2.807°$, so $\theta_{ss} = 87.19°$. **Steady-state error ≈ 2.81° (the arm sags below the target).** This is small because $K_p$ is much larger than $mg\ell$ ($K_p/mg\ell \approx 20$).

**(d)** Gravity compensation reduces the steady-state error to about 0.5°, not exactly zero, since the compensation is evaluated at the measured angle and the sensor and model are not perfect.

Poles: unchanged. The linearisation of the gravity term about $\theta_d$ is $mg\ell \cos\theta_d$, which is zero at 90°, so the stiffness stays $K_p$.

## Question 3

*Given: $A = [0, 1; 0, -2]$, $B = [0; 1]$, $C = [1, 0]$, $f_s = 1$ kHz so the sample time is $T = 1/f_s = 1$ ms.*

**(a)** Reason 1 – noise amplification: the backward difference $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequency ($\approx \omega$ for small $\omega T$), i.e. it is a

high-pass operation. Any noice on the encoder signal therefore appears in the velocity multiplied by roughly $1/T = 1000\ s^{-1}$, and can swamp the true velocity.

Reason 2 – quantisation: the encoder only reports integer counts, so $\Delta\theta$ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder $q = 2\pi/4096 = 1.53\times10^{-3}$ rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

**(b)** The analogue prototype is the first-order low-pass $H(s) = \omega_c/(s + \omega_c)$ with $\omega_c = 2\pi f_c = 2\pi \times 20 = 125.66$ rad/s. The bilinear (Tustin) transform replaces s by $(2/T)(z - 1)/(z + 1)$ with $T = 1/f_s = 0.001$ s.

Define $a = \omega_c T/2 = 125.66 \times 0.001/2 = 0.062832$. Then $H(z) = \omega_c / [(2/T)(z-1)/(z+1) + \omega_c] = a(z + 1) / [(1 + a) z - (1 - a)] = a(1 + z^{-1}) / [(1 + a) - (1 - a) z^{-1}]$.

Dividing through by (1 + a): $a/(1 + a) = 0.062832/1.062832 = 0.059117$ and $(1 - a)/(1 + a) = 0.937168/1.062832 = 0.881767$. Hence with u = differenced signal and y = filtered velocity: $\mathbf{y_k = 0.05912\ (u_k + u_{k-1}) + 0.88177\ y_{k-1}}$. (Frequency pre-warping is neglected since $f_c \ll f_s/2$.)

Verification of the DC gain: set u and y constant (z = 1): $y = 2(0.05912)u + 0.88177\ y \Rightarrow y(1 - 0.88177) = 0.11824\ u \Rightarrow y/u = 0.11824/0.11823 = 1.000$. In general $H(1) = 2a/[(1 + a) - (1 - a)] = 2a/2a = 1$ exactly, so the steady-state output equals the commanded input ✓.

**(c)** First check observability: O = [C; CA] = [1, 0; 0, 1], full rank, so the observer poles can be placed arbitrarily.

With $L = [\ell_1, \ell_2]^T$ and C = [1, 0], LC = $[\ell_1, 0; \ell_2, 0]$, so $A - LC = [-\ell_1, 1; -\ell_2, -2]$.

Characteristic polynomial: $\det(sI - (A - LC)) = \det[s + \ell_1, -1; \ell_2, s + 2] = (s + \ell_1)(s + 2) - (-1)(\ell_2) = s^2 + (2 + \ell_1)\ s + (2\ell_1 + \ell_2)$.

Both estimator pols at s = −10 means the desired polynomial is $(s + 10)^2 = s^2 + 20\ s + 100$. Comparing coefficients: $s^1$: $2 + \ell_1 = 20 \Rightarrow \ell_1 = 18$. $s^0$: $2\ell_1 + \ell_2 = 100 \Rightarrow \ell_2 = 100 - 36 = 64$.

$\mathbf{L = [18, 64]^T}$. Verification: A − LC = [−18, 1; −64, −2] has trace −20 and determinant (−18)(−2) − (1)(−64) = 36 + 64 = 100, so the polynomial is $s^2 + 20s + 100 = (s + 10)^2$ ✓.

**(d)** Advantage of the observer that no fliter of this kind can give: the observer is driven by both the measurement y and the model with the actual input u, so it predicts the motion between samples. The velocity estimate therefore has little or no phase lag. The low-pass filter on differenced data, by contrast, always pays a lag of about $1/(2\pi f_c) = 8$ ms for its noise rejection, a trade-off it cannot escape. In addition the observer gives all states, and can be augmented with a disturbance state (e.g. unknown load torque) which no filter on θ can estimate.

Circumstance where the filter is safer: when the plant model is poorly known or changes during operation, for example an unknown payload, varying friction, contact with the environment, or actuator saturation making u differ from the commanded value. The observer propagates the wrong model so its estimate can be biased (and in the worst case unstable with the feedback loop), whereas the model-free filter just smooths what the encoder actually measures.

*Note: observer poles are usually placed several times faster than the controller poles.*

## Question 4

*Given: plant $\ddot{\theta} + a\dot{\theta} + c\theta = b\ u$ with a = 2, c = 6, b = 1; controller $u = N_{ff}\ r - k_1\theta - k_2\dot{\theta}$.*

**(a)** Substitute the control law into the plant: $\ddot{\theta} + a\dot{\theta} + c\theta = b\ N_{ff}\ r - b\ k_1\ \theta - b\ k_2\ \dot{\theta}$. Moving the feedback terms to the left: $\ddot{\theta} + (a + b\ k_2)\ \dot{\theta} + (c + b\ k_1)\ \theta = b\ N_{ff}\ r$.

Numerically, $a + b\,k_2 = 2 + 2 = 4$ and $c + b\,k_1 = 6 + 2 = 8$, so the characteristic equation is $s^2 + 4s + 8 = 0$. Roots: $s = [-4 \pm \sqrt{(16 - 32)}]/2 = -2 \pm 2j$. They have negative real parts, so the loop is stabel; $\omega_n = \sqrt{8} = 2.83$ rad/s and $\zeta = 4/(2 \times 2.83) = 0.707$.

Transfer function from r to θ: $T(s) = b\,N_{ff} / (s^2 + 4s + 8)$. DC gain ($s = 0$): $T(0) = N_{ff}/8$. For unity DC gain, $\mathbf{N_{ff} = 8}$. (Note that this is larger than 1 because the gravity-related stiffness c and the feedback $k_1$ both resist the displacement.)

**(b)** With the actual $b = 0.75$ and unchanged $N_{ff} = 8$, $k_1 = k_2 = 2$, the closed loop becomes $\ddot{\theta} + (a + bk_2)\dot{\theta} + (c + bk_1)\theta = bN_{ff}\,r$, i.e. $\ddot{\theta} + 3.5\dot{\theta} + 7.5\theta = 6\,r$.

For a unit step the steady state is found by setting derivatives to zero: $7.5\,\theta_{ss} = 6 \times 1$, so $\theta_{ss} = 6/7.5 = \mathbf{0.8}$.

Tracking error $= (r - \theta_{ss})/r \times 100\% = (1 - 0.8) \times 100 = \mathbf{20\%}$. A common mistake would be to say the output falls by the same 25% as the actuator gain; that is wrong here, because b also appears in the feedback stiffness $c + bk_1$, which decreases from 8 to 7.5 and partially compensates. The poles also change: $s^2 + 3.5s + 7.5 \rightarrow s = -1.75 \pm 2.11j$, still stable.

**(c)** Take the state $z = [\theta, \dot{\theta}, \xi]^T$. The plant gives $\ddot{\theta} = -a\dot{\theta} - c\theta + b\,u$ with $u = -k_1\theta - k_2\dot{\theta} + K_I\xi$, and $\dot{\xi} = r - \theta$. With $r = 0$ (r only shifts the equilibrium) the closed-loop matrix is $A_{cl} = [0, 1, 0; -(c + bk_1), -(a + bk_2), bK_I; -1, 0, 0]$.

Characteristic polynomial $\det(sI - A_{cl})$, expanding along the first row: $s\cdot[(s + a + bk_2)s - 0] + 1\cdot[(c + bk_1)s + bK_I] = s^3 + (a + bk_2)s^2 + (c + bk_1)s + bK_I$. **(ReusIt)**

Desired poles $-4$ and $-2 \pm 2j$: $(s + 2 - 2j)(s + 2 + 2j) = (s + 2)^2 + 4 = s^2 + 4s + 8$. Multiply by $(s + 4)$: $s^3 + 4s^2 + 8s + 4s^2 + 16s + 32 = s^3 + 8s^2 + 24s + 32$.

Compare coefficients with $a = 2$, $b = 1$, $c = 6$: $s^2$: $2 + k_2 = 8 \Rightarrow \mathbf{k_2 = 6}$. $s^1$: $6 + k_1 = 24 \Rightarrow \mathbf{k_1 = 18}$. $s^0$: $b\,K_I = 32 \Rightarrow \mathbf{K_I = 32}$.

Check: all coefficients of the desired polynomial are positive and $(8)(24) = 192 > 32$, so Routh–Hurwitz confirms the three poles are in the left half plane ✓.

**(d)** Suppose the closed loop is stable and r is constant. Then the state converges to a constant, so $\dot{\theta} = \ddot{\theta} = 0$ and, crucially, $\dot{\xi} = 0$ as well (ξ itself is constant at its final value $\xi_{ss}$). From the integrator equation $\dot{\xi} = r - \theta$, we get $r - \theta_{ss} = 0$, i.e. $\boldsymbol{\theta_{ss} = r}$.

Neither b nor c appears in this argument. They only influence the value of $\xi_{ss}$ needed to hold the equilibrium (from the plant equation $c\theta_{ss} = bu_{ss} = b(-k_1\theta_{ss} + K_I\xi_{ss})$), and the pole locations, so the zero error survives any error in the model parameters provided the loop stays stable (unlike the feed-forward $N_{ff}$ scheme of part (b)).

Practical failure mode: integrator windup. If the actuator saturates (torque limit), the loop is effectively open, the error stays non-zero and ξ keeps integrating to a huge value. When the reference becomes reachable the stored integral forces a large overshoot and a long recovery. Standard remedy: anti-windup, such as clamping the integrator when the actuator is saturated, conditional integration, or back-calculation that feeds the difference between commanded and saturated torque back to the integrator.